Lecture 12

16 February 2026 19:08

Recap

Minima of convex functions ($S \subseteq \mathbb{R}^n$ convex, $f: S \to \mathbb{R}$ convex):

$$\min_{x \in S} f(x)$$

- If x^* is a local minimum then it also a global minimum.
- If f is ~~strictly convex~~ then there is a unique minimum.
- (S ~~open)~~ x^* is a minimum $\Leftrightarrow$ ~~$\nabla f(x^*)$~~ $= 0$.
  - If $\bar{x}$ is not a global minimum, it cannot be a local minimum (last time)
  - Proof Assume x_1 and x_2 are two minima.

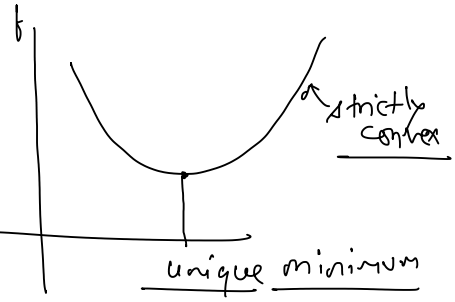
    $\Rightarrow f(x_1) = f(x_2)$

    $f(\underbrace{\tfrac{1}{2}x_1 + \tfrac{1}{2}x_2}_{\in S}) < \tfrac{1}{2}f(x_1) + \tfrac{1}{2}f(x_2)$

    $= f(x_1)$

    $\Rightarrow \frac{x_1 + x_2}{2}$ yields a smaller value than x_1

    $\Rightarrow x_1$ cannot be a minimum (contradiction)



  - we already know that $\nabla f(x^*)$ is a necessary condition for optimality of x^* from convexity of f,

    for $x \in S$ $\quad f(x) \geq f(x^*) + \underbrace{\nabla f(x^*)}_{=0}{}^T (x - x^*)$ ✓

So, if $\nabla f(x^*) = 0$

$\Rightarrow f(x) \geq f(x^*) \quad \forall x \in S$

$\Rightarrow x^*$ is a minimum (global)

Maxima of concave functions ($S \subseteq \mathbb{R}^n$ convex, $f: S \to \mathbb{R}$ concave):

[Equation]

- If x^* is a local maximum then it also a global maximum.
- If f is strictly concave then there is a unique maximum.
- (S open) x^* is a maximum $\Leftrightarrow \nabla f(x^*) = 0$.

Examples

(1) $f(x) = \|x\|^2$

$\min_{x \in \mathbb{R}^2} f(x)$

f is convex.

$\|x\|^2 = x_1^2 + x_2^2$

$\nabla f(x) = \begin{bmatrix} 2x_1 \\ 2x_2 \end{bmatrix}$

$\nabla^2 f(x) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ (PD $\forall x \in \mathbb{R}^2$)

$\Rightarrow f$ is strictly convex.

$\Rightarrow$ there is unique minimum

$\begin{bmatrix} 2x_1 \\ 2x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\nabla f(x) = 0 \Rightarrow x^* = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

② $f: S \longrightarrow \mathbb{R}$

$S = \{x \in \mathbb{R}^2 : x_1 > 0,\ x_2 > 0,\ x_1 + x_2 < 1\}$

$f(x) = -\ln(1 - x_1 - x_2) - \ln x_1 - \ln x_2$

$\min_{x \in S} f(x)$

$$\nabla f(x) = \begin{bmatrix} \frac{1}{1-x_1-x_2} - \frac{1}{x_1} \\ \frac{1}{1-x_1-x_2} - \frac{1}{x_2} \end{bmatrix}$$

A is PD $\Leftrightarrow x^T A x > 0 \ \forall x \neq 0$

& A symmetric

$$D^2 f(x) = \begin{bmatrix} \frac{1}{(1-x_1-x_2)^2} + \frac{1}{x_1^2} & \frac{1}{(1-x_1-x_2)^2} \\ \frac{1}{(1-x_1-x_2)^2} & \frac{1}{(1-x_1-x_2)^2} + \frac{1}{x_2^2} \end{bmatrix}$$

$d \in \mathbb{R}^2$

$$d^T D^2 f(x) d = [d_1\ d_2] \begin{bmatrix} & \\ & \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$$

$$= \frac{1}{(1-x_1-x_2)^2}(d_1 + d_2)^2 + \frac{d_1^2}{x_1^2} + \frac{d_2^2}{x_2^2} > 0 \quad \forall d \neq 0$$

$\Rightarrow D^2 f(x)$ is PD $\forall x \in S$

$\Rightarrow f$ is strictly convex.

$\Rightarrow$ unique minimum

$$\nabla f(x) = 0 \Rightarrow \begin{bmatrix} \frac{1}{1-x_1-x_2} - \frac{1}{x_1} \\ \frac{1}{1-x_1-x_2} - \frac{1}{x_2} \end{bmatrix} = 0$$

$\Rightarrow x_1 = x_2 = 1/3$

$\Rightarrow x^* = \begin{bmatrix} 1/3 \\ 1/3 \end{bmatrix}$ is the unique solution

Lipschitz Continuity

Lipschitz continuity (a set [Equation] , a function [Equation]): [Equation] is said to be Lipschitz continuous with parameter [Equation] if

[Equation] for all [Equation]

- If [Equation] is Lipschitz continuous then it is also continuous on [Equation] .

-otherway need not hold.

Example ① $f: [0, 10] \longrightarrow \mathbb{R}$

$f(x) = x^2$ is Lipschitz continuous with parameter 20.

Easy check: $|x^2 - y^2| \leq 20\,|x - y| \quad \forall x, y \in [0, 10]$

② $f: \mathbb{R} \longrightarrow \mathbb{R}$ $f(x) = x^2$ continuous but not Lipschitz continuous

first order characterization

$f: S \longrightarrow \mathbb{R}$ is Lipschitz ctz with parameter L and is differentiable $\Longleftrightarrow \|\nabla f(x)\| \leq L$

$f: S \longrightarrow \mathbb{R}$ differentiable ($S \subseteq \mathbb{R}^n$)

$\nabla f(x) : S \longrightarrow \mathbb{R}^n$

$\nabla f(x)$ Lipschitz ct. with parameter L

$\|\nabla f(x) - \nabla f(y)\| \le L\|x-y\| \quad \forall x, y \in S$

(f is also called L-smooth)

$S \subseteq \mathbb{R}^n$, [Equation] continuously differentiable: [Equation] is said to have a Lipschitz continuous gradient with parameter [Equation] iff

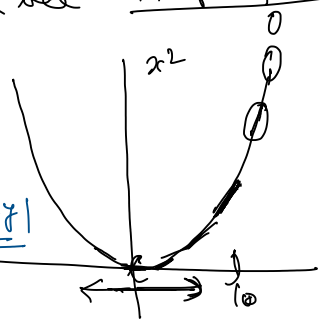
[Equation] for all [Equation]

- Second order ~~characterization~~ (~~[Equation] is twice~~ differentiable)

  [Equation] all [Equation]

If f is L-smooth ($\nabla f(x)$ is Lipschitz ct. with parameter L)

then $\|\nabla^2 f(x)\, d\| \le L\|d\|$ (see notes for proof)



L-smooth

$|f'(x) - f'(y)| \le L|x-y|$

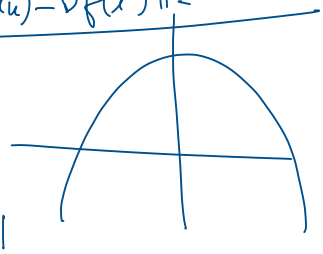
$\min_{x \in S} f(x)$

$\min x^*$

$\nabla f(x^*) = 0$

$\|\nabla f(x_k)\| \approx 0$

$\|\nabla f(x_k) - \nabla f(x^*)\| \le L\|x - x^*\|$



$d^T \nabla^2 f(x)\, d \le \|d\|\, \|\nabla^2 f(x)\, d\|$

$\le L\|d\|^2 = L d^T I_n d$

$\Rightarrow d^T(\nabla^2 f(x) - LI)d \le 0 \quad \forall d$

$\Rightarrow \nabla^2 f(x) - LI$ is NSD

$\Rightarrow \lambda_{max}(\nabla^2 f(x)) \le L$

(2nd order characterization)

$\boxed{\lambda(A - \lambda I) = \lambda(A) - \lambda}$

$\lambda(\nabla^2 f(x) - LI) = \lambda(\nabla^2 f(x)) - L$

f is strongly convex with parameter m

$\Rightarrow \nabla^2 f(x) - mI$ is PSD $\quad \forall x$

$\Rightarrow \lambda_{min}(\nabla^2 f(x)) \ge m \quad \forall m$

- If [Equation] is strongly convex with parameter [Equation] and has Lipschitz continuous gradient with parameter [Equation] then [Equation] .

$\frac{L}{m}$: condition number

≥ 1

Taylor's Series.

$f(y) = f(x) + \nabla f(x)^T (y-x) + \frac{1}{2}(y-x)^T \nabla f(\alpha x + (1-\alpha)y)(y-x)$ for $\alpha \in [0,1]$

$\leq \frac{1}{2} L \|y-x\|^2$

$\Rightarrow f(y) \leq f(x) + \nabla f(x)^T (y-x) + \frac{L}{2}\|y-x\|^2$

Descent Lemma ([Equation] , [Equation] continuously differentiable): If [Equation] has a Lipschitz continuous gradient with parameter [Equation] then [Equation] for all [Equation]

Gradient Algorithms

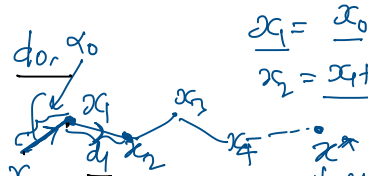
$\min_{x \in S} f(x)$ $S \subseteq \mathbb{R}^n$

- necessary cond. for (local) optimality $\nabla f(x) = 0$ (we cannot solve this directly)

Iterative Algorithms

$x_1 = x_0 + \alpha_0 d_0$ (scalar > 0)

$x_2 = x_1 + \alpha_1 d_1$



where $\nabla f(x^*) = 0$

$f(x_0) \geq f(x_1)$
$f(x_1) \geq f(x_2)$
$f(x_2) \ldots$

directions: $d_0, d_1, d_2, \ldots$ $\|d_i\| = 1 \ \forall i$

step sizes: $\alpha_0, \alpha_1, \alpha_2, \ldots > 0$

we aspire to have $\lim_{t \to \infty} x_t = x^*$ such that $\nabla f(x^*) = 0$

(x^* need not be an minimum, may not even be a local minimum.

- however it is a global minimum if f is convex)

Unconstrained optimization: [Equation]
[Equation]
Goal: Obtain a [Equation] such that [Equation]
Initialize: [Equation]
Iterate: for [Equation] where [Equation] and [Equation]

- [Equation] is called a step-size
- [Equation] is called a direction
- The algorithm is called a descent algorithm if [Equation] for all [Equation]

$$\left(\begin{bmatrix} x_{t+1,1} \\ \vdots \\ x_{t+1,n} \end{bmatrix} = \begin{bmatrix} x_{t,1} \\ \\ x_{t,n} \end{bmatrix} + \alpha_t \begin{bmatrix} d_{t,1} \\ \vdots \\ d_{t,n} \end{bmatrix} \right)$$

$x_{t+1} - x_t = \alpha_t d_t$

$\|x_{t+1} - x_t\| = \alpha_t \underbrace{\|d_t\|}_{=1}$

$\|d_t\|_2 = \sqrt{\sum_i d_{t,i}^2} = 1$

$\|(1 \ \ldots 1)\| = \sqrt{1+1} = 1$

$\left\|\left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)\right\| = 1$

direction selection

Taylor's series first order approximation

$$f(x_{t+1}) = f(x_t) + \nabla f(x_t)^T (x_{t+1} - x_t) + (x_{t+1} - x_t)^T r_1 (x_{t+1} - x_t)$$

$$= f(x_t) + \alpha_t \nabla f(x_t)^T d_t + \alpha_t d_t^T r_1 (\alpha_t d_t)$$

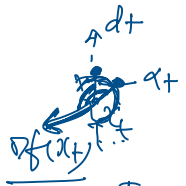
$\alpha_t \ll 1 \Rightarrow$ negligible compared to the other terms

To make $f(x_{t+1}) < f(x_t)$

we should choose d_t such that

$\nabla f(x_t)^T d_t < 0$ (this guarantees descent only if α_t is small enough)

choice of d_t depends on the gradient



$\nabla f(x_t)^T d_t =$

$$f(x_{t+1}) - f(x_t) = \alpha_t \underbrace{\nabla f(x_t)^T d_t}_{<0}$$

we would like $|\nabla f(x)^T d_t|$ to be large

$|a^T b| = \|a\| \|b\|$ if $b = \pm \alpha a$

$$|\nabla f(x)^T d_t| \le \|\nabla f(x_t)\| \underbrace{\|d_t\|}_{=1} = \|\nabla f(x_t)\|$$

For equality, we should use $d_t = -\frac{\nabla f(x_t)}{\|\nabla f(x_t)\|}$

(this gives maximum descent)

Direction selection:

- If [Equation] is chosen such that [Equation] for all [Equation] then the algorithm is called a gradient descent algorithm
- If [Equation] for all [Equation] then the algorithm is called a steepest descent algorithm

$$x_{t+1} = x_t + \alpha_t \left(-\frac{\nabla f(x_t)}{\|\nabla f(x_t)\|}\right)$$

$$= x_t - \frac{\alpha_t}{\|\nabla f(x_t)\|} \nabla f(x_t)$$

(this guarantees descent only if α_t small)

$x_{t+1} = x_t + \alpha_t d_t$ where $\|d_t\| = 1$

Step-size selection:

- Fixed step-size: [Equation]

$x_{t+1} = x_t + \alpha d_t$ where d_t is such that $d_t^T \nabla f(x_t) < 0$

If $d_t = -\frac{\nabla f(x_t)}{\|\nabla f(x_t)\|}$ (if $\nabla f(x_t) = 0$ then no need to go to x_{t+1})

$x_{t+1} = x_t - \frac{\alpha}{\|\nabla f(x_t)\|} \nabla f(x_t)$ is not called fixed step size

$\alpha_t = \alpha \|\nabla f(x_t)\|$ is called fixed stepsize.

$\|x_{t+1} - x_t\| = \alpha \frac{\|\nabla f(x_t)\|}{\|\nabla f(x_t)\|}$

- if [Equation] then, customarily, fixed step-size refers to [Equation] for all [Equation]

$x_{t+1} = x_t - \alpha \nabla f(x_t)$

$\|x_{t+1} - x_t\| = \alpha \|\nabla f(x_t)\|$ not fixed for all t

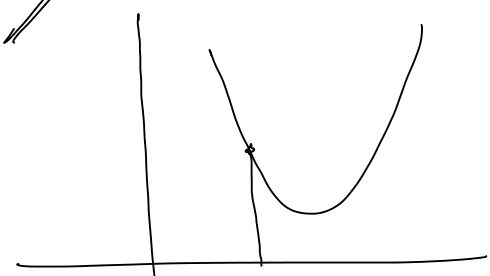
if $\nabla f(x_t) = 0$ then

$x_{t+1} = x_t - \frac{\alpha}{\|\nabla f(x_t)\|} \cdot \nabla f(x_t)$

$\nabla f(x_t) = 0$

$= 0$

$\Rightarrow x_{t+1} = x_t$



Next time: Problem with fixed step size!